Task 3 – Wireshark Packet Analysis

Alfido Tech Cybersecurity Internship

Submitted By

Name: Eeshan Garg

Candidate ID: BS/REG/120082

Role: Cybersecurity Intern

Organization: Alfido Tech

Task: Task 3 – Wireshark Packet Analysis

Tools Used: Wireshark, TCPDump

Date: May 2026

1. Objective

The objective of this task was to analyze multiple packet capture (PCAP) files using Wireshark and perform network forensic investigations. The analysis focused on identifying protocols, suspicious activities, attacker behavior, compromised hosts, and data exfiltration attempts through packet inspection and protocol analysis.

This task helped in understanding:

- Network traffic analysis
- Packet inspection
- Protocol filtering
- Incident response techniques
- Threat detection
- Digital forensics methodology

2. Tools Used

The following tools were used during the investigation:

- Wireshark
- TCPDump
- Ubuntu Linux Environment

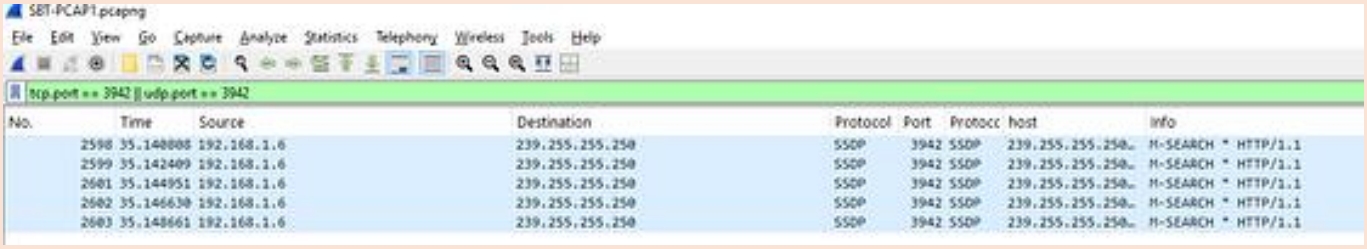
3. PCAP 1 – Network Fundamentals

This analysis focused on identifying baseline traffic and host activity using display filters and endpoint statistics.

Q1. Which protocol was used over port 3942?

Query

tcp.port == 3942 || udp.port == 3942



Answer

SSDP

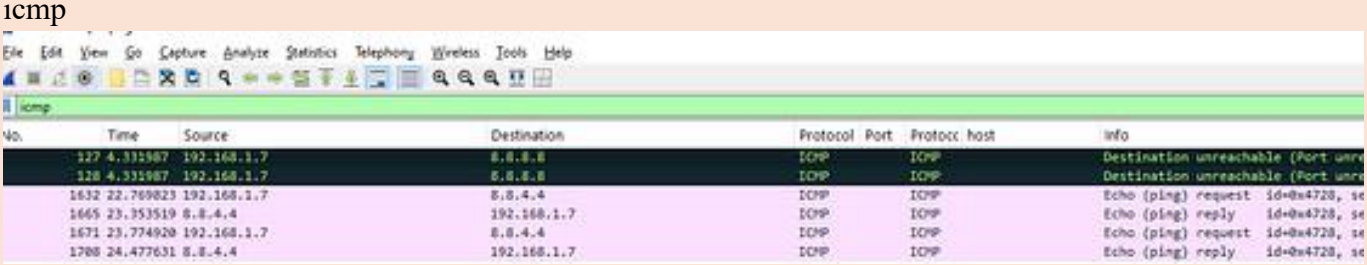
Reason

Filtering traffic for port 3942 revealed packets associated with the Simple Service Discovery Protocol (SSDP).

Q2. What is the IP address of the host that was pinged twice?

Query

icmp



Answer

8.8.4.4

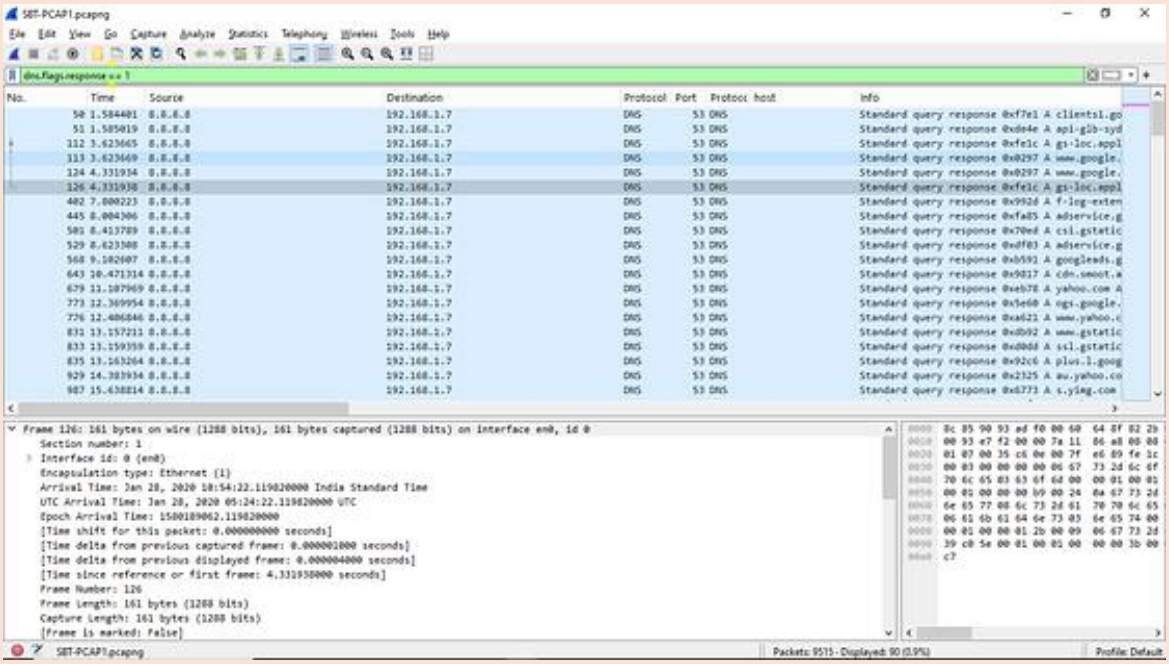
Reason

The ICMP filter displayed two Echo Request packets sent to 8.8.4.4.

Q3. How many DNS query response packets were captured?

Query

dns.flags.response == 1



Answer

90

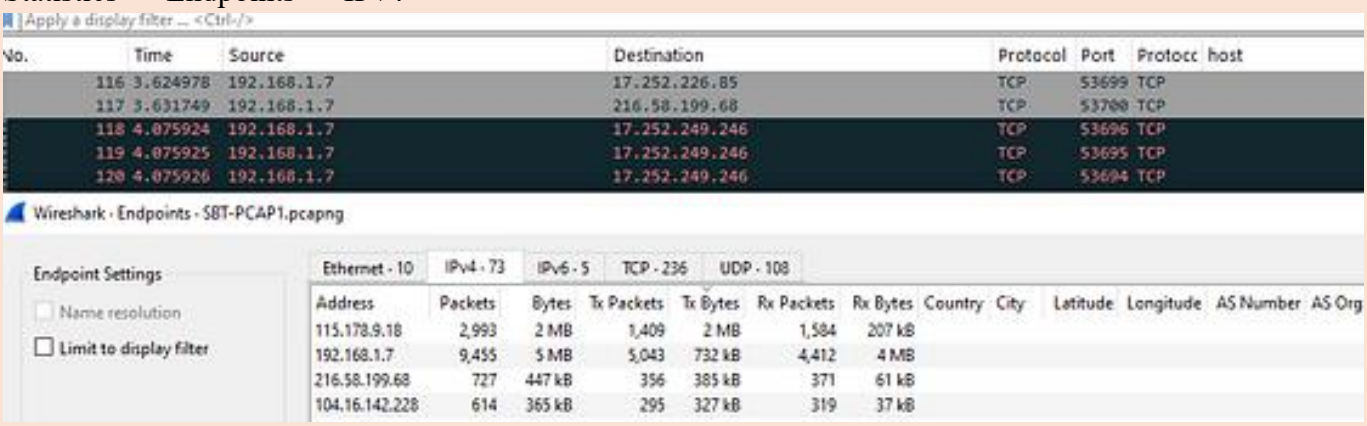
Reason

Applying the DNS response filter displayed a total of 90 DNS response packets.

Q4. Which host transmitted the highest amount of data?

Step

Statistics → Endpoints → IPv4



No.	Time	Source	Destination	Protocol	Port	Protocc host
116	3.624978	192.168.1.7	17.252.226.85	TCP	53699	TCP
117	3.631749	192.168.1.7	216.58.199.68	TCP	53700	TCP
118	4.075924	192.168.1.7	17.252.249.246	TCP	53696	TCP
119	4.075925	192.168.1.7	17.252.249.246	TCP	53695	TCP
120	4.075926	192.168.1.7	17.252.249.246	TCP	53694	TCP

Address	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	Latitude	Longitude	AS Number	AS Org
115.178.9.18	2,993	2 MB	1,409	2 MB	1,584	207 kB						
192.168.1.7	9,455	5 MB	5,043	732 kB	4,412	4 MB						
216.58.199.68	727	447 kB	356	385 kB	371	61 kB						
104.16.142.228	614	365 kB	295	327 kB	319	37 kB						

Answer

115.178.9.18

Reason

Sorting by transmitted bytes identified this host as the top data sender.

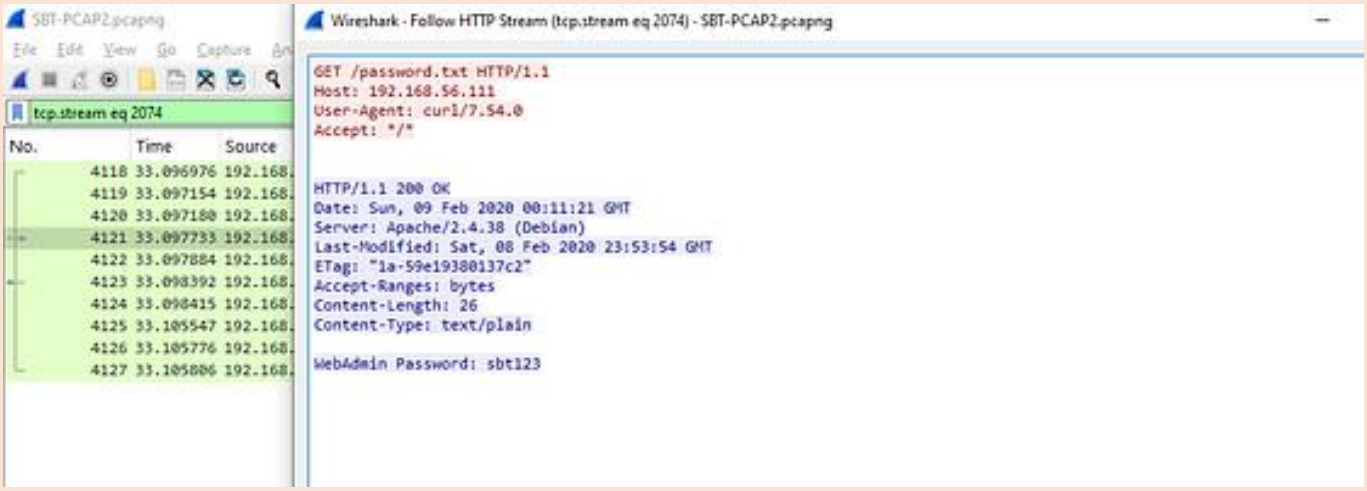
4. PCAP 2 – Compromised Windows Host

This investigation focused on analyzing a compromised Windows system to identify attacker activity, credentials, and accessed files.

Q1. What is the WebAdmin password?

Query

frame contains "password"



Answer

sbt123

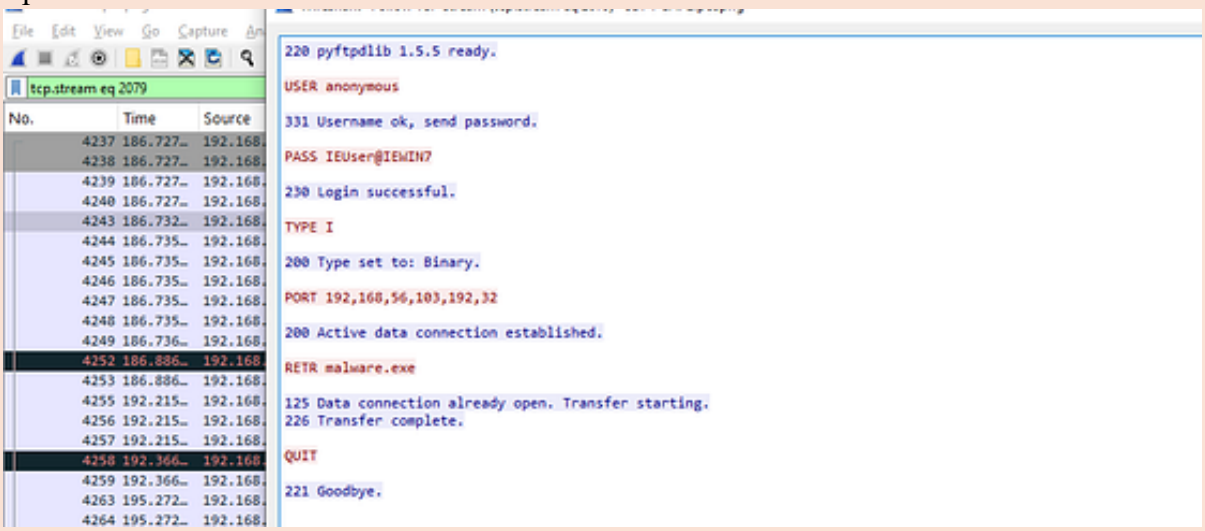
Reason

The password was identified by following the HTTP stream.

Q2. What is the version number of the attacker's FTP server?

Query

ftp



Answer

1.5.5

Q3. Which port was used to gain access to the victim host?

Query

tcp.port == 8081

Answer

8081

Reason

Following the TCP stream revealed an interactive command session running through port 8081.

Q4. What was the name of the confidential file discovered?

Answer

Employee_Information_CONFIDENTIAL.txt

Reason

The file name was identified from the directory listing within the TCP stream.

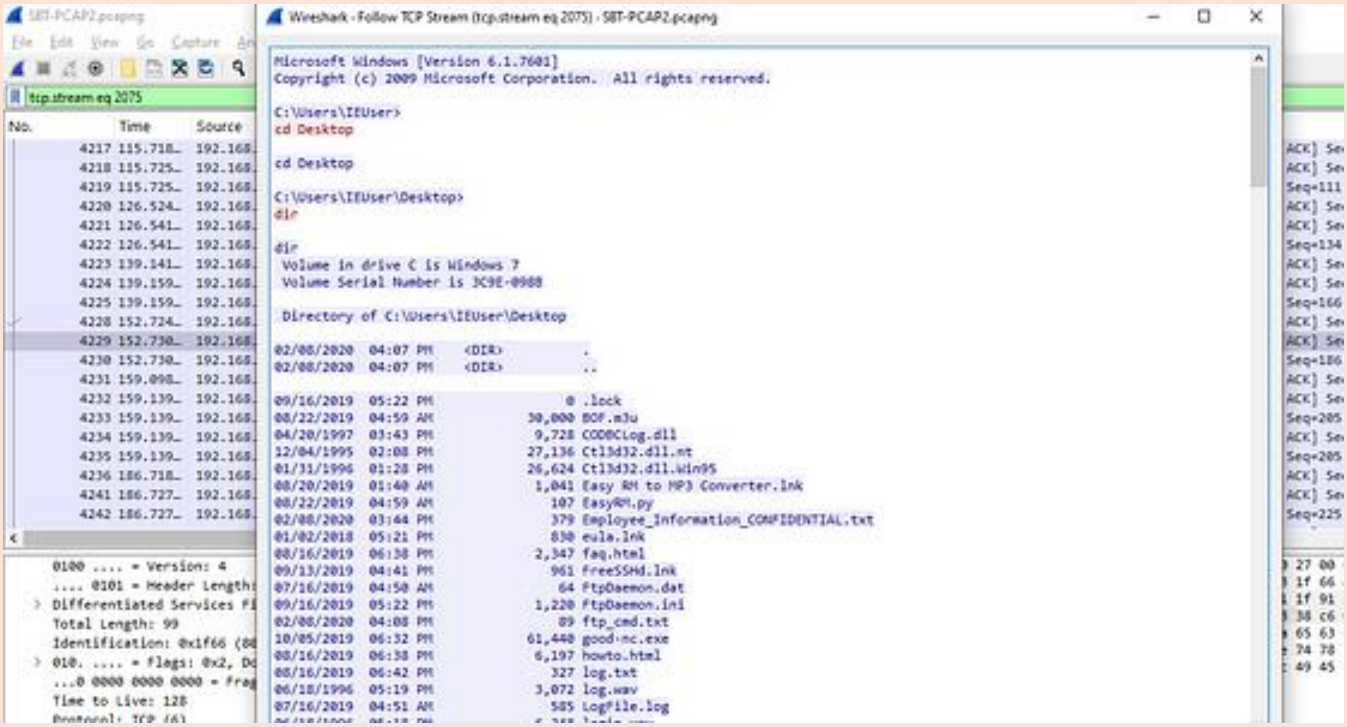
Q5. Which log file was created at 4:51 AM?

Answer

LogFile.log

Reason

The file timestamp was identified through directory listing analysis.



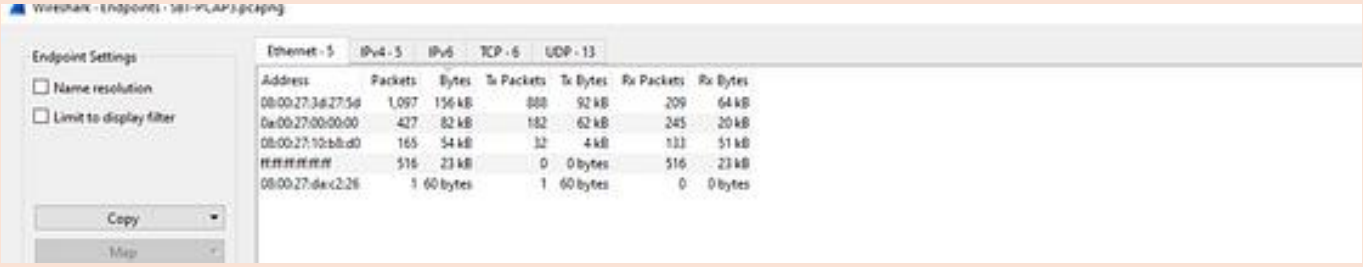
5. PCAP 3 – Man-in-the-Middle Attack & Data Sniffing

This investigation focused on detecting malicious interception of internal communications and identifying attacker activity.

Q1. What is the MAC address of the attacker?

Step

Statistics → Endpoints → Ethernet



Answer

08:00:27:3d:27:5d

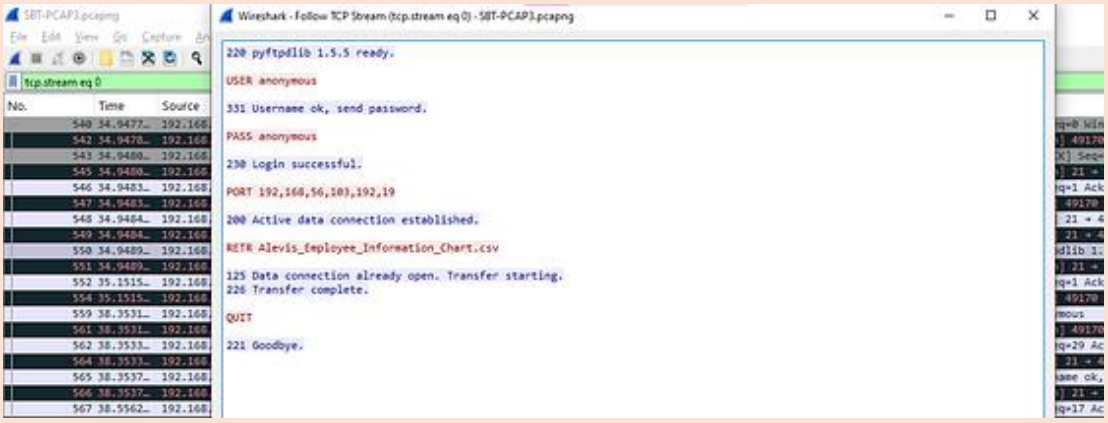
Reason

The MAC address was identified as the most active forwarding device in the traffic capture.

Q2. Which file was downloaded from the central server?

Query

ftp



Answer

Alevis_Employee_Information_Chart.csv

Reason

The file transfer was identified by following the FTP stream.

Q3. What type of attack was identified?

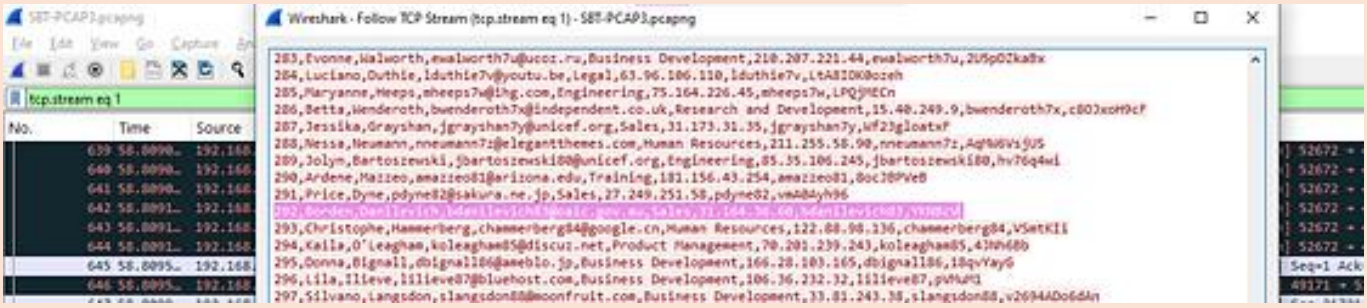
Answer

ARP Poisoning / ARP Spoofing

Q4. Which department did Borden Danilevich work in?

Query

frame contains "Borden"



Answer

Sales

Q5. What was the SSH password of the Domain Administrator?

Answer

YKNbcV

Reason

The password was identified within the exfiltrated CSV data.

6. Key Findings

The investigation successfully identified:

- Protocol-specific traffic
- DNS responses
- ICMP activity
- FTP transfers
- SMB communications
- ARP spoofing activity
- Credential exposure
- Data exfiltration indicators
- Service banners
- Packet volumetrics

The analysis demonstrated how packet inspection can reveal attacker behavior, compromised systems, and sensitive information leakage.

7. Conclusion

This task demonstrated the effectiveness of Wireshark in network traffic analysis and digital forensics investigations. Through packet filtering, protocol analysis, and stream reconstruction, multiple attack scenarios and suspicious activities were successfully identified.

The exercise improved practical understanding of:

- Network forensics
- Incident response
- Packet analysis
- Threat detection
- Protocol investigation
- Traffic monitoring

The task provided valuable hands-on exposure to real-world cybersecurity investigation techniques using Wireshark.